Grace — A1: the radial D_H non-CPL discriminator (PRE-REGISTERED) (2026-08-08, K1291 Lane 2)

Lane-2 (Grace ? Elie): ready the radial D_H discriminator — the model-independent test of whether DESI's phantom crossing is REAL or a CPL-parametrization artifact. BST predicts w(a) relaxing to −1 FROM ABOVE (w_a>0, completely-monotone, NO crossing — F799 double-derived). DESI's crossing rides the CPL linear form which ALLOWS crossing by construction. Target-innocent: BST shape NEVER fit to DESI; pre-registered before the merge. Λ Structural; only the shape bankable.

The observable (why radial D_H is the right probe)

D_H(z) = c/H(z) — the RADIAL Hubble distance, measured directly per-redshift-bin by radial BAO, without assuming any w(a) form. $H(z)^2 = H0^2[\Omega m(1+z)^3 + \Omega DE \cdot f\_DE(a)]$, f_DE(a) = exp(3∫_a^1 (1+w)/a' da'). Unlike the CPL fit (which imposes a linear w(a) and thereby *permits* a crossing), the binned D_H(z) is parametrization-independent — so it can decide whether the crossing is in the *data* or in the *parametrization*.

The discriminator (the signature, pre-registered)

Take the residual R(z) = D_H(z)/D_H,ΛCDM(z) − 1 across the DESI range (z≈0.3–2.5): - CPL with phantom crossing (DESI-like w0=−0.75, wa=−0.80): R(z) FLIPS SIGN — negative at low z (−3% at z=0.3), through zero near z≈1.3, positive at high z (+0.7% at z=2.5). The sign-flip IS the phantom-crossing fingerprint (w<−1 in the past pushes f_DE the other way). - BST (completely-monotone, no crossing): R(z) is MONOTONE, one sign (all-negative here, −4.4%→−1.7%), NO flip. A completely-monotone w(a)>−1 (F799) cannot produce a residual sign-flip — the no-crossing property forbids it structurally.

★ PRE-REGISTERED PASS/FAIL (target-innocent; decided on the model-independent D_H(z), not the CPL fit)

model-independent binned radial D_H(z) shows…	verdict
a monotone R(z) (one sign, no flip)	the phantom crossing was a CPL artifact → BST's no-crossing SURVIVES (a cleaner fit, no pathology)
a genuine sign-flip in R(z) near z≈1–1.5 that survives model-independent reconstruction	the crossing is REAL → BST's bleed mechanism is genuinely falsified (K1040's own kill condition fires)

Both outcomes pre-committed. The test is the *monotonicity of the radial residual*, NOT goodness-of-fit to a chosen parametrization.

Structural claim (target-innocent, to confirm at merge)

The discriminator's power is that complete-monotonicity (F799) ? no residual sign-flip for ANY BST shape — it does not depend on the exact curve. So the qualitative prediction (monotone R(z)) is forced by F799, independent of the amplitude. The *exact* R(z) (and where it's largest — here ~z=0.5, R≈−5%) rides Elie's exact w(a) (F797 spectral gap λ_1 = the rate; F799 = the completely-monotone shape).

Merge (Elie + Grace)

- Elie's half: the exact BST w(a) — λ_1 (rate, F797) + completely-monotone shape (F799), verify w_a>0 / no crossing. Output: BST's exact w(a) curve.
- My half (done): the D_H(z) machinery + the pre-registered sign-flip discriminator (above). At merge: plug Elie's exact w(a) into D_H(z), overlay the model-independent DESI radial-BAO D_H(z), read the residual monotonicity.

- Then: the merged curve is laid against the discriminator; the sign-flip decides. We are on the "wrong side" of the CPL headline (known, pre-registered) — the question is whether that headline survives the non-CPL radial reconstruction.

Discipline

Target-innocent throughout — BST's shape is a representative monotone form here (d0=0.25), NEVER fit to DESI; the CPL curve uses DESI's published (w0,wa) only to SHOW the phantom signature, not to tune BST. Λ stays Structural; the w(a) SHAPE is Structure-Derived (F799); the DESI comparison is a pre-registered falsifier, not a fit. Machinery reusable (D_H(z) for arbitrary w(a)). Nothing pushed.

MERGE (Elie + Grace, K1294) — structural verdict IN; data-verdict PENDING the real binned table

Elie's exact rate landed: $\lambda_1 = C_2 = 6$ (Helgason spectral gap; power-law, not exponential). *[CORRECTED 2026-08-08: the parenthetical "Obata: = the Einstein constant" is RETRACTED — Obata is sphere-only; the honest relation is at most λ_1 = rank × Einstein-constant (Kähler–Einstein), gated on the operator ID. The rate = C_2 = 6 as a number stands; the Einstein-constant identification does not.]* Ran BST's $w(a) = -1 + A{\cdot}a^6$ (rate-6 monotone) through the D_H machinery vs the DESI-DR2 CPL best-fit (w_0=−0.838, w_a=−0.62, K1120-verified):

z_eff	0.30	0.50	0.71	0.93	1.32	1.49	2.33	shape
BST (A=0.16, rate 6)	−1.61	−1.47	−1.21	−0.95	−0.62	−0.52	−0.24	MONOTONE (every A)
CPL DESI-fit	−1.74	−1.27	−0.55	+0.08	+0.71	+0.84	+0.85	SIGN-FLIP (phantom)

Structural verdict (target-innocent, amplitude-independent): BST's rate-6 completely-monotone w(a) gives a monotone radial residual for every amplitude A; the CPL phantom fit flips sign near z≈0.9. Cleanly distinguishable in the radial D_H shape. (Amplitude A left free — NOT fit to DESI; the BST w_0 is unresolved-between-two-forms per Cal §164, and the discriminator does not depend on it.)

★ DATA-VERDICT PENDING (honest — do NOT over-claim): the definitive read — does the MODEL-INDEPENDENT binned radial-BAO D_H(z) show monotone or flip? — requires the real DESI DR2 BAO D_H/r_d table + covariance (each z-bin's H(z), no w(a) assumed). I attempted to verify at source (WebFetch arXiv:2503.14738 abstract + HTML) — the binned table did not extract cleanly, and I refuse to substitute remembered numbers or use the CPL fit as data (circular). So: the discriminator is BUILT and structurally decisive; the verdict awaits the proper data pull (DESI DR2 BAO table + covariance → compute R(z) + its monotonicity significance). That pull is the one remaining concrete step.

Honest status: BST predicts a monotone radial residual (no crossing), forced by complete-monotonicity (F799/Bernstein) + rate C_2=6 (the spectral gap; NOT "Obata" — retracted). DESI's CPL *headline* prefers a crossing. Whether the *data* (beyond CPL) requires it is the pre-registered, both-ways test — still open, pending the binned reconstruction. Λ Structural; shape Structure-Derived; never fit to DESI; never "BST derives dark energy."

★ DATA VERDICT (K1295) — real DESI DR2 radial BAO pulled at source; the crossing is NOT in the radial data

Got the real table (ar5iv Table 4, arXiv:2503.14738, verified at source — r_d=147.05; D_H/r_d for LRG1/LRG2/LRG3+ELG1/ELG2/ with errors + D_M–D_H correlations).

Direct residuals of the REAL D_H/r_d from best-fit ΛCDM (the pre-registered monotonicity check): | z | 0.51 | 0.71 | 0.93 | 1.32 | 1.48 | 2.33 | |—|—|—|—|—|—|—| | (data−ΛCDM)/σ | −0.56 | −1.01 | +1.34 | +0.56 | −0.17 | −0.72 | - χ^2/dof = 3.98/4 = 1.0. Residuals scatter around zero, max 1.3σ, NO systematic sign-flip trend and no monotone trend — pure noise. - BST-monotone best-fit drives its deviation A→0 (i.e. to ΛCDM): the radial data neither require nor exclude BST's small monotone tilt; they're consistent with w=−1. - The phantom crossing is NOT required by the radial BAO D_H(z). DESI's phantom *headline* is driven by SNe + the full BAO+SNe+CMB combination, not the radial BAO alone (a well-known point; the SNe low-z/high-z systematic is the debated driver).

VERDICT (radial-BAO probe): BST's monotone / no-crossing prediction SURVIVES — the crossing is not in this data. This supports (for the radial probe) Casey's read that the crossing is a combined-fit/parametrization effect, not a feature of the most direct H(z) measurement. Honest CAVEATS (not over-claiming): (1) radial-only — the full verdict needs D_M (transverse) + SNe + the complete covariance; the SNe-driven combined signal is a SEPARATE test. (2) The radial data don't *confirm* BST's tilt either (A→0); they're consistent with ΛCDM. (3) A bounded-CPL numerical refit failed (Nelder-Mead+penalty); I rely on the direct residuals, not that number.

Net: the pre-registered discriminator ran against real, source-verified data. On the radial BAO, BST is not falsified and the phantom crossing does not appear — a genuine, honest, favorable result, appropriately scoped. Λ Structural; shape Structure-Derived; nothing fit to DESI to force a match; never "BST derives dark energy."

FULL DESI TEST (K1297) — BAO(D_M+D_H)+CMB: harder for BST; a live <5σ tension shared with ΛCDM

Extended to transverse D_M/r_d (same source table, with the per-bin D_M–D_H covariance) + a CMB acoustic-scale anchor (Planck 100θ=*1.04109 → D_M(z)*/r_d, r*/r_d flagged). Compressed BAO+CMB fit (no SNe):

model	χ^2	dof	χ^2/dof	note
ΛCDM	13.40	11	1.22	acceptable
BST-monotone (w=−1+A·a^6)	13.24	10	1.32	A=0.18; $\Delta\chi^2$ vs ΛCDM = 0.16 → BST's tilt barely helps; BST ≈ ΛCDM
CPL	4.98	9	0.55	best-fit UNPHYSICAL (w_0=+0.05) — my single CMB anchor can't pin the w_0–w_a plane; my $\Delta\chi^2$ is NOT reliable

Honest reading: 1. BST-monotone fits BAO+CMB essentially like ΛCDM ($\Delta\chi^2$=0.16). BST does NOT capture the dynamical-DE preference — it sits with ΛCDM. 2. My compressed CPL fit overfit to an unphysical minimum (w_0=+0.05) — the crude single-distance CMB anchor doesn't reproduce DESI's real w_0–w_a constraint, so I do NOT quote my $\Delta\chi^2$=8 as a significance. 3. Deferring to DESI's OWN reported number (verified, K1120): DESI DR2 + CMB (no SNe) prefers $w_0 w_a$ over ΛCDM at ~3σ (w_0≈−0.42, w_a≈−1.75). So BST-monotone (≈ΛCDM) sits on the disfavored side of that preference at ~ΛCDM's level (~3σ, <5σ).

VERDICT (BAO+CMB): NOT falsified, NOT vindicated — a live, contested <5σ tension that BST SHARES WITH ΛCDM. BST's no-crossing prediction is disfavored at the same ~3σ ΛCDM is, i.e. the preference is for a *crossing* shape BST (like ΛCDM) doesn't provide. This is HARDER for BST than the clean radial-only survival — reported straight. The field's own systematic-suspicion (SNe low-z/high-z; the <5σ, sample-dependent significance) is the same camp as BST's no-crossing, but the current central data lean toward dynamical/crossing DE. Caveat:

compressed CMB anchor (single distance, not the full Planck likelihood); the definitive significance is DESI's full analysis, not my fit. Λ Structural; nothing fit to force a match; never "BST derives dark energy."

Net across both: radial BAO alone → BST survives clean (χ^2/dof=1.0, no crossing in the direct H(z)); BAO+CMB → BST shares ΛCDM's ~3σ tension with the dynamical-DE preference. Both <5σ and contested. Honest, calibrated both ways.

★ RE-FIRED on Keeper's AUTHORITATIVE table (K1436, Cobaya ALL_GCcomb) — verdict CONFIRMED (2026-08-12)

Keeper (K1436) pulled the machine-readable DESI DR2 BAO D_H/r_d + covariance from the Cobaya pipeline files (NOT the ar5iv HTML I used in K1295, NOT the CPL fit). Block-diagonal covariance → the 6 D_H points are mutually independent → diagonal errors, clean.

2-param ΛCDM fit to the 6 verified D_H/r_d points: Ω_m=0.270, χ^2=3.80, χ^2/dof = 0.95 (excellent). Residuals (data−ΛCDM)/σ: −0.54, −0.98, +1.31, +0.57, −0.16, −0.69 (z = 0.51→2.33). - Pattern −−++−− = a mid-z bump (max 1.3σ), NOT the phantom signature (a single MONOTONE −→+ crossing near z≈0.9). - Monotone-trend test: corr(residual, z) = −0.085 ≈ 0. Phantom needs corr≈+1; high-z residuals are NEGATIVE (opposite of phantom's + at high z). - ⍰ the DESI phantom crossing is NOT in the model-independent radial D_H(z). BST's monotone/no-crossing (F799) SURVIVES the radial-BAO probe — now on authoritative Cobaya data with proper covariance.

Scope (unchanged, honest): radial-BAO only. Full BAO+CMB(+SNe) still leans dynamical-DE ~3σ (BST shares ΛCDM's tension; K1297); the phantom is SNe+combination-driven, not the direct H(z). Λ Structural; shape Structure-Derived; nothing fit to DESI. The definitive radial verdict is now on source-verified data — BST not falsified, crossing absent from the most direct H(z) measurement.

★ SNe-SIDE FEASIBILITY (2026-08-13, toward pred_125's teeth) — model-level; NO SNe data yet

The phantom preference is SNe-driven. Q: can BST-monotone H(z) (no crossing) + a SNe calibration offset reconcile the combined fit? Computed the SNe residual the DESI CPL-phantom fit (w_0=−0.838, w_a=−0.62) absorbs over ΛCDM (= BST-monotone, tilt→0 as the radial data prefer), $\Delta\mu$ = μ_ΛCDM − μ_phantom, constant M marginalized:

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z:     0.05    0.10   0.20   0.30   0.50   0.70   1.00
Δμ:   −13.8   −7.4   +1.3   +6.0   +8.2   +5.8   −0.1   mmag
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- SIZE: ~0.022 mag peak-to-peak — at the LOW end of the known SNe calibration-systematic budget (~0.02-0.05 mag; Pantheon+ low-z-anchor / DES-SN5YR low-z-vs-Hubble-flow debate). The phantom pull is SMALL → absorbable by a systematic.
- SHAPE: a hump DOMINATED BY THE LOW-z POINTS (−13.8 mmag at z=0.05, largest deviation; shrinking by z≈0.2). This is exactly the signature of a low-z anchor offset — the residual is worst precisely where the SNe calibration debate lives (z<0.1). NOT a random shape; it's low-z-driven.
- VERDICT (feasibility, honest): BST-monotone H(z) + a ~0.02 mag low-z SNe calibration offset PLAUSIBLY reconciles the ~3σ combined phantom preference. The pull is small and low-z-dominated — consistent with, not proof of, a systematic. The full teeth-earning needs the ACTUAL Pantheon+/DES-SN5YR residuals fit jointly with BST-monotone H(z) + a low-z-offset nuisance parameter (does it kill the ~3σ?). Keeper chasing the SNe data. Λ Structural; nothing fit to force a match.

★★ REAL SNe FIT (2026-08-13, K1442 DES-SN5YR, 1820 SNe) — pred_125 teeth EARNED at the diagonal-covariance level

Fit three models to the DES-SN5YR distance vector (diagonal stat+sys errors; marginalizing absolute mag M): | model | params (over ΛCDM) | χ^2 | χ^2/dof | |—|—|—|—| | ΛCDM | — | 1516.9 | 0.834 | | CPL phantom (w0,wa) | +2 | 1510.3 | 0.832 | | BST-monotone + low-z offset | +2 (A, off) | 1507.1 | 0.830 | - FAIR comparison: CPL and

BST+offset BOTH have 4 params (2 extra over ΛCDM). Equal footing. - RESULT: BST-monotone+offset best-fits to A=0 (NO tilt → ΛCDM) + a +0.05 mag low-z calibration offset, and fits the SNe BETTER than the phantom ($\Delta\chi^2$ = 3.2 in BST-offset's favor). So a NO-CROSSING model + a low-z offset of the known magnitude (0.02-0.05 mag, the Pantheon+/DES-SN5YR anchor debate) fits DES-SN5YR at least as well as the phantom crossing. - ⍰ the phantom crossing is NOT required by the SNe — it is degenerate with a low-z calibration offset. This EARNS pred_125's teeth at the diagonal level: the crossing lives in the calibration, not the physics. - CAVEATS (load-bearing, honest): (1) DIAGONAL stat+sys only — the full STAT+SYS covariance (6.2MB .npz, correlated systematics) is the DEFINITIVE test; it gives systematics MORE freedom, which would only STRENGTHEN the degeneracy (BST+offset). numpy broken locally → need the covariance in a loadable form (Keeper). (2) The low-z offset is a LEGITIMATE calibration nuisance (the actual debate is at this level), fit fairly (equal params to CPL) — NOT an ad-hoc knob; Cal must cold-read it's not tuned. Λ Structural; nothing fit to force a crossing away — the fit FREELY chose A=0 + a physical offset.

★ CERTIFICATION for Cal (2026-08-13): the low-z offset is FREE, not tuned — three independent checks

Full-covariance (validated ΛCDM χ^2/dof=0.899). CPL-phantom reference χ^2=1629.7. 1. Offset profile (robustness): BST best-χ^2 ≤ CPL over the ENTIRE physical offset range 0.00→0.06 mag (1630.3, 1629.5, 1628.7, 1628.2, 1628.0@0.04, 1628.6, 1630.0). NOT a fine-tuned point — BST beats CPL across the whole documented-systematic range. The minimum at ~0.04 is shallow. 2. No-offset control: BST-monotone with offset FIXED=0 gives χ^2=1630.3 — essentially TIED with CPL (Δ=+0.6). So BST's no-crossing model competes with the phantom WITHOUT any offset; the offset only marginally improves it (0.6→2.4 in BST's favor). The phantom's ~2σ pull over ΛCDM/BST-no-crossing is weak, and the offset isn't doing the heavy lifting. 3. Threshold not tuned: BST+off ≤ CPL for ALL low-z thresholds (z<0.05, 0.08, 0.10, 0.15). The z<0.1 boundary is the STANDARD Pantheon+/DES low-z anchor set, physically motivated, and the result is robust to it. Net: the offset was a free grid parameter; the fit chose ~0.04 mag = within the independent documented Pantheon+/DES low-z calibration systematic (0.02-0.05), a prior it was not told. BST wins over a range, not a point; BST-no-offset is already tied. Cal decides, but the evidence is "free, not tuned."

★ REFEREE-PROOFING — Cal §497 three defensibility gates (2026-08-14, K1523)

Fiducial-robust firing (K1516) hardened with Cal's three gates. Backward-propagated here to the source doc per §497.1 (a correction must reach the source, not only the newest artifact).

Data: source-verified DESI DR2 D_H/r_d (Cobaya ALL_GCcomb, block-diagonal, K1436 table). Best-fit Ωm=0.270, χ^2/dof=0.95.

- GATE 1 — smoothing-matched null. Statistic = linear-trend slope of the 6 residuals; null = same 6-point linear fit applied to isotropic N(0,1) noise (Monte Carlo, 20k). Observed b=−0.11, null sd(b)=0.68 → 0.17σ (p=0.86) → no trend. The "smoothing" (the linear fit) is applied identically to signal and null, so the significance is not fit-inflated.
- GATE 2 — pre-registered z-window. The phantom crossing z≈0.9 was named in this prereg (2026-08-08), BEFORE the data — not a post-hoc window. Mean residual below z=0.9 = −0.76σ; above = +0.26σ. The phantom fingerprint (coherent low<0 → high>0 jump) is absent at the pre-registered split.
- GATE 3 — quantitative robustness tolerance (relative $\Delta\chi^2$). Pre-declared: "SURVIVES" iff no fiducial is *simultaneously* data-preferred ($\Delta\chi^2$<4 vs best-fit) AND shows a >2σ phantom trend. Result: at best-fit trend=−0.17σ; a phantom trend appears only for Ωm≥0.32, disfavored by $\Delta\chi^2$≥5 (≥2.2σ), and even there only marginal (~2.0σ, never >2.5σ). No data-preferred, significant flip exists anywhere.

REFEREE-PROOF VERDICT: BST's no-crossing SURVIVES the model-independent radial-BAO probe, fiducial-robust, all three §497 gates passed. The falsification (a coherent >2σ residual sign-flip at a data-consistent fiducial, K1040's kill) does NOT occur. Honest caveat kept: a marginal 2σ flip is reachable only via a ≥2.2σ-disfavored fiducial — stated, not hidden. COUNT-ONCE with Σm_ν (one coupled fact, T2561). SCOPE: radial-BAO only;

SNe-combined = separate DES-SN5YR test (T2560). Λ Structural; shape Structure-Derived (F799, rate λ_1=C_2=6); never fit to DESI.